

LC 1870: Minimum Speed to Arrive on Time (PhonePe)

Authored by: **Arpit Raj**, LNMIIT Jaipur

Date: July 15, 2026

Problem Overview

I/P -> a dist vector, train leave only on ith hour

dist vector has distance in km that ith train rides, all trains are taken in sequential order

O/P -> min speed of trains

Logic & Approach

if speed x works, $>x$ works too,

hence

-> (monotonic + min) -> Binary search

since the trains leave at only integer time we might have to wait between trains

So for every segment we use `ceil(time)` (Except last)

last train will use actual time **

Code Implementation Thought Process

hence

```
for (int i = 0 ; i < n-1 ; i++ ) {  
    totalTime += ceil( (double) dist[i] / sp );  
}
```

for every suspected given speed, we find how much time would it take.

Last train doesn't need waiting as it drops directly to office and man doesn't need to wait for any next train

```
totalTime += (double) dist[n-1] / sp;
```

C++ Solution

Below is the complete implementation based on the discussed logic:

```
class Solution {
public:
    int n;
    bool check(int speed, vector<int>& dist, double hour){
        double t=0;
        for(int i=0; i<n-1; i++){
            t+=ceil((double)dist[i]/speed);
        }

        t+=(double)dist[n-1]/speed;
        if(t>hour){
            return false;
        }
        return true;
    }
    int minSpeedOnTime(vector<int>& dist, double hour) {
        n=dist.size();

        int l=1; //l cant be 0 otherwise time=d/s=infinite , not possible
        int h=100000000; //given in question

        int ans=-1; //default maan ke chalo nhi phuch paayega

        while(l<=h){
            int mid=l+(h-l)/2;

            if(check(mid, dist, hour)){
                ans=mid;
                h=mid-1;
            }

            else{
                l=mid+1;
            }
        }
        return ans;
    }
};
```

